Interview Notes: Call with Denny

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Bottom line from Denny: **weather is the single biggest variable in predicting ship movement**, so we should keep it — but as a separate layer we can toggle, not baked into the historical training data.

Who and why

- **Interviewee:** Denny, retired U.S. Navy captain who ran the civil maritime intelligence analysis department at the Office of Naval Intelligence after 9/11. His team built the well-known global AIS "ship traffic" map.
- **Purpose:** sanity-check our ship-tracking / route-prediction approach and settle whether to include weather.
- **Call length:** about 19 minutes.
- **Citing him:** he said we can use him as a source and name him as a retired Navy captain from ONI. His surname came through unclearly on the recording (sounds like "Bradley"), so confirm spelling before it goes in anything we submit.
- **Note:** these notes come from an auto-transcript of the call, so wording is paraphrased.

Key takeaways

1. **Keep weather, as a layer.** Weather is the most common reason ships leave their normal routes. Treat it as its own layer we can turn on and off. In historical training data it "complicates" the model, so isolate it rather than mixing it in.
2. **Weather has to be live during a voyage.** For predicting an ongoing trip, weather needs constant real-time updates. Free shipping and weather data sources exist.
3. **Ships follow known sea lanes for a reason.** Routes are the fastest, cheapest, safest way to move cargo on very thin margins. A deviation from the usual route is the signal worth flagging.
4. **Pick a region.** He recommends a focused area; New York / New Jersey came up as our likely choice.
5. **Stick to commercial vessels.** Warships don't behave like commercial ships. Skip cargo tracking for now — it's much harder.
6. **AIS data is messy.** Ships go dark, spoof positions, and smaller or domestic vessels may not broadcast at all. Plan for gaps and bad points.
7. **There's a commercial story.** Shipping companies and supply-chain players running just-in-time inventory need exactly this kind of prediction.
8. **Judges care about who we talked to.** Real expert interviews could separate us from other teams, like last year.

Detailed notes

How AIS works

- AIS (Automatic Identification System) is required by the International Maritime Organization for ships above a certain size on international voyages, under the SOLAS (Safety of Life at Sea) treaty.
- It was built for **collision avoidance**: a line-of-sight ship-to-ship broadcast of identity, heading and speed. Tracking ships with it came later.
- After 9/11, satellites started picking up AIS from space, which made global tracking possible.
- Each ship is identified by its **MMSI** number.
- Denny said ships broadcast roughly every few seconds (he recalled about every 4).
- U.S.-only or smaller ships may not be required to broadcast, so choose a minimum ship size for our dataset.

Data quality problems

- **Going dark:** ships can switch AIS off. Off the Horn of Africa, ships turned it off so Somali pirates couldn't find them.
- **Spoofing:** bad actors fake positions or IDs. One spoofed ship showed up about 200 nautical miles inland.
- **Coverage gaps:** glitches, non-compliance and missing receivers leave holes. His team crunched a full year of AIS data (about 265,000 ships) on a supercomputer to map where coverage was good and where it was empty. He thinks AI could now do that in seconds.

Predicting movement

- **Dead reckoning:** estimate position from a ship's last known course, speed and characteristics (size, weight, power). Different ships behave differently.
- **Sea lanes:** known, efficient routes. Historical data shows them clearly.
- **"Hockey sticks":** when lots of ships suddenly veer the same odd way, it's almost always weather, such as a hurricane in the Gulf.
- Ship types matter: bulk carriers vs. container ships. Tracking cargo on top of ships is a more advanced problem.

Why ships deviate

Cause	How common / how to treat it
Weather	Most common by far; model as a separate, live layer
Engine failure	Occasional anomaly
Crew arrested in port	Occasional anomaly
Insurance problems	Occasional anomaly
Political events	Occasional anomaly
Nefarious activity	Rare; a route that "makes no financial sense" is the tell

Anomaly detection examples

- **Ferry scenario:** a France–UK ferry runs the same route on a schedule. If it leaves that route and weather doesn't explain it, you can infer something is wrong (e.g. a hijacking where the crew can't call for help).
- **Illogical routes:** going around the Horn of Africa instead of through the Suez Canal, or taking a dangerous route for no reason, gave bad actors away.
- His watch floor had meteorologists on duty full time because weather affected everything.

Commercial value

- Global supply chains run on **just-in-time delivery**: companies keep little inventory and plan manufacturing around arrivals.
- Margins are very thin, and an idle ship makes no money, so shipping and logistics companies need accurate position and arrival predictions.

Notable quotes

Lightly cleaned from the auto-transcript, so check against the recording before quoting him word for word.

- "The weather is the biggest variable."
- "Ships go from point A to B to C in the fastest, cheapest, safest possible way to make money."
- "If it's floating around and doesn't have cargo or isn't going somewhere, it's not making money."
- On historical weather data: "I would isolate that and maybe put that in a layer that you could turn on and off."
- On catching bad actors: "This doesn't make financial sense — no one would do that."
- "The people involved in the global supply chain need this data … they have very thin margins."

What this means for the build

- ☐ **Weather as a toggleable layer:** keep it separate from the core route model, with a switch in the UI and the pipeline.
- ☐ **Clean training data:** flag or filter voyages that were bent by major weather events (e.g. hurricanes), so the baseline route model learns normal sea lanes.
- ☐ **Live weather feed** for in-progress voyage predictions.
- ☐ **Scope the data:** NY/NJ region, commercial vessels only (drop military), minimum vessel size.
- ☐ **Data cleaning:** key on MMSI; drop impossible positions such as points on land; handle AIS gaps (ship goes dark) without breaking tracks.
- ☐ **Baseline prediction:** dead reckoning from last course and speed, adjusted by vessel size and type.
- ☐ **Anomaly flag:** deviation from a vessel's usual route that weather doesn't explain.
- ☐ **Optional extra:** a coverage-gap map showing where AIS data is thin.
- ☐ **Out of scope for now:** cargo tracking.

Data sources

- U.S. historical AIS data going back to around 2009 is free. He may have meant NOAA/BOEM's MarineCadastre.gov (unclear on the recording).
- Denny is sending a list of free shipping and weather data sources, plus a written analysis he texted Pierce.

Follow-ups

- ☐ Pierce: forward Denny's document and data-source list to the group.
- ☐ Confirm Denny's full name and title for citations.
- ☐ Reach out to the second contact he suggested: an offshore oil and gas expert (name unclear on the recording, sounds like "John Holland"). A Houston / Gulf angle would add hurricane and weather-anomaly cases.
- ☐ Denny offered to talk to the rest of the team anytime.